

The Possibilities of Using Virtual Environments in Research on Wayfinding

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Abstract

The study evaluated the degree of comparability of human behaviour exhibited during an evacuation scenario conducted in two environments, one real and the other virtual. For this purpose, we created a precise 3D model (digital twin) of a real campus building. In both the experimental (virtual environment) and control (real environment) groups, the participants were all first-time visitors to the building. The groups were instructed to locate a target room where they would subsequently be required to complete a simple task, at which point an evacuation alarm would be set off. We evaluated three types of data collected during their quest to find a way out of the building: an eye-tracker logged gaze hits on various objects (e.g., navigation signs), recorded the locomotion and trajectory of the participants, and logged a combination of these parameters (e.g., points along the path where participants sighted certain objects). The sample was composed of 73 participants (35 for the real environment, 38 for the virtual environment). We analysed and confirmed the results and concluded that despite certain aspects of human behaviour differing during the simulated evacuation, the locomotion behaviour exhibited by the participants in both environments was generally comparable. To improve the potential use of these findings in practice, we selected easily accessible hardware and excluded expensive devices such as treadmills, etc.

1. Introduction

Evacuation behaviour is an important and complex area of study. For simulation purposes, real-life mass evacuations would be unethical to create, therefore many research studies are done retrospectively from data containing insufficient information. Experiments in real buildings are highly susceptible to interruption since the possibility of being able to seal off and place an entire building under controlled conditions is very low. A solution to this is the use of a virtual environment (VE). Before the use of a VE can be recommended for this purpose, it is crucial to explore and evaluate the comparability in human behaviour under the conditions of real environments (REs) and virtual environments; this is the main aim of the current study.

Differences in the perception of REs and VEs have been observed and recorded. In VEs, participants often perceive distances as shorter (Renner, Velichovsky & Helmert, 2013), colours as too bright, and time appears contracted (Kuliga et al., 2015, Mullen & Davidenko, 2021). Generally, VEs have been described as too artificial, empty and sterile. This perceived shortcoming could partly be a consequence of the intentional plainness of experimental studies in which corridors and rooms have minimal or no furnishings (lack of windows, decorations, etc.). In more recent studies, details have been added to improve the realism of environments and thus also the ecological validity of the study (Wiener et al., 2020). Still, very few researchers have compared a virtual environment which is identical to a known real environment, therefore the full extent of possibilities is unknown. However, some studies which compared evacuation behaviour in real and virtual environments have indeed demonstrated high ecological validity (Feng, Duives & Hoogendoorn, 2021). Virtual environments seem to have little to no effect on spatial performance. (Varmaghani et al., 2022; Pastel et al., 2022). If the special aspects of immersive virtual reality are addressed, it could be a promising tool for behavioural research in the future (Kuliga et al., 2015).

Various aspects of human behaviour must be tracked and measured for a valid comparison of these environments. Eye-tracking glasses headsets can be used to pinpoint significant visual cues. Previous research has found that some evacuation signs are improperly placed and remain entirely unnoticed during evacuation (Bae et al., 2020). Environmental characteristics also affect decision-making. The more light there is in a corridor, the greater the likelihood a person will use it for evacuation (Hidayetoglu, Yildirim & Akalin, 2012; Vilar et al., 2013). Another decision point during evacuation is whether to use the elevator (Mossberg, Nilsson & Andréé, 2020). The technology for tracking locomotion in real environments still varies greatly. Automated geolocation is currently too demanding to apply, and camera systems are frequently used instead (Feng, Duives & Hoogendoorn, 2021). Automated location tracking in a VE, however, is relatively easily implemented (Ugwitz et al., 2019, Luoto, 2018).

Navigation behaviour in an unfamiliar environment is referred to as wayfinding and has four main strategies: piloting (navigation from landmark to landmark), exploration, recognition of distant landmarks, and use of celestial compasses (Golledge et al., 2000). Hölscher et al. (2006) proposed three wayfinding strategies for multi-level buildings: central point strategy (following well-known paths), direction strategy (leading directly to the horizontal position of the destination), and floor strategy (finding the desired floor first). Snopková et al. (2021) discussed the relationship between the space-syntax parameters of a building and the choice of retracing strategy adopted during an evacuation scenario.

In the current study, we evaluated the extent to which navigation behaviour in virtual and real environments is comparable. Unlike many previous studies, we compared both eye-tracking data and locomotion data to pinpoint the most helpful orientation points during an evacuation. Finally, we linked these two data sets to identify the positions of the first hits on areas of interest (AOIs).

For the experiment, we created a realistic virtual 3D model (i.e., a digital twin of an existing university building). Participants were allocated into either a control group for the real environment or an experimental group for the virtual environment. All participants completed a standardised task. First, they searched through an unfamiliar building. A fire alarm was then activated, and the participants were required to evacuate the building. We recorded their movements through the building in addition to their eye-movements.

2. Methods

As mentioned above, we selected detailed parameters to investigate the differences and similarities between behavioural patterns during an evacuation. Specifically, these parameters were personal trajectory, gaze hits on relevant objects (e.g., navigation signs, other unrelated signs, furniture, windows, etc.), and a combination of both (e.g., distance from where a particular object was sighted). We examined the similarity and dissimilarity measures for each of these parameters.

Since the study's aim was to provide evidence-based instructions for designing future evacuation simulations, we used commonly accessible technology to compare the two environments. For this purpose, we describe a detailed procedure which takes into account evacuation, safety and fire protection and can be

easily applied in practice for building design. The procedure includes considerations for optimal data collection conditions, with a focus on mainly human factors (hydration of participants, maximum time spent in the VE, occurrence of motion sickness, etc.) and effective implementation of technology.

2.1 Experimental Environments

The experiment was performed in a university building (FAST VUT) in Brno, Czech Republic. The late Baroque style building (B) has six floors, with a layout in the shape of a squared figure of eight. It is connected to the main building (A) by a covered footbridge on the third floor. Access to this footbridge was barred for the purposes of the experiment. The evacuation route led from the corner classroom on the fourth floor, along corridors to the central staircase, and then out to the front of building B (Fig. 1). Access to the classroom was also possible through a different entry and via the elevator to the fourth floor; this allowed us to partially observe the spatial knowledge acquired by participants on the way to the corner classroom (the starting point for evacuation). Access on the fourth floor of the building was easily altered by barring the access corridor and blocking the route which participants already knew from the way in.

Recreated in 3D, the virtual twin of the building used existing BIM (building information modelling) schematics generated in Autodesk. We converted these schematics into the Unity real-time graphics engine using the Tridify conversion service. Although the converted model retained correct spatial proportions, some problems remained from the conversion process; many of the imported 3D shapes and objects were not optimised for real-time rendering, containing either too many polygons, erroneous geometry, missing textures or unfitting representations. These objects therefore required re-creation using 3D modelling software (Blender). The graphic artist also had to visit the physical site to capture the existing textures in the building and transfer these details to the virtual environment. Similarly, capturing the realistic feel of indoor lighting required processing with pre-baked ray tracing and considered placement of light sources (Fig. 2). This rather time-consuming process is described in detail in a work by Kvarda (2020). As a result of the procedure, the final VE may be considered realistic (See Virtual scene realism factor, Poesch & Döring, 2013).

To ensure the experiment ran smoothly, we incorporated interactive elements such as text instructions presented along the way, audio (e.g., the alarm) and dynamic objects into the environment. Dynamic objects included invisible blocks which restricted users from moving beyond the intended path, barred doorways which forced the user to seek a different route, and doors/elevators which executed their classic revolving/transporting functionality. All of these elements were added to the Unity environment using a series of C# scripts.

Eye-tracking and performance optimizations required additional enhancements in the environment (Ugwitz et al, 2022); virtual objects were named and partitioned correctly, and the AOIs they resembled were in concordance with the data extracted from the experiment's real environment.

2.2 Hardware

In the RE, we used two cameras. The first camera was a set of SensoMotoric Instruments (SMI) Eye Tracking Glasses 2 (sampling rate at 60 Hz, video resolution at 960 x 720p, 30 FPS). The glasses were connected to an external battery for power and a mobile phone for data storage and control. The second camera was a Forever SC-210 Plus action cam (1920 x 1080p, 30 FPS) carried around by the researcher who followed behind the participants along their evacuation route.

In the VE, we used an HTC Vive Pro Eye headset. The virtual scene was rendered at its native resolution of 1440 x 1600 per eye, at 90 Hz. The integrated eye tracker collected data at 120 Hz. We used Lighthouse Position Tracking system (gen 2), which is native to HTC Vive headsets. When the user fixated or remained stationary during virtual exploration and movement using keyboard arrow keys, the positional tracking system allowed the participant to tilt and glance around and retain full semblance of bodily mobility. The virtual scene was rendered on a PC configured with an AMD Ryzen 7 3700X CPU, nVidia RTX2070 Super GPU, 64GB DDR4 RAM, NVMe SSD, and Windows 10 OS.

2.3 Software

The system made use of multiple software packages. Revit 2020 and Tridify cloud service (Tridify Ltd., 2020) provided conversion of the building information model into Unity 2018.4.22f1, where it was further optimised. Smaller scale interior and exterior models were created in Blender 2.80. The executable build of the virtual application was compiled according to the project created in the aforementioned Unity engine. To facilitate eye-tracking functionality, the project was extended with Vive SRanipal SDK 1.3. The set of scripts in Toggle Toolkit was used to log user-environment interactivity and participant behaviour (Ugwitz et al., 2021).

SMI BeGaze 3.7 was used to analyse the eye-tracking data collected from the SMI eye-tracking glasses. Additional statistical analysis was performed in ArcGIS Pro 2.7.0, QGIS 3.10, GRASS GIS 7.8.4, RStudio v. 1.4.1106 and PostgreSQL 10.11 database with PostGIS extension. Visualisations were created in ArcGIS Pro 2.7.0, ggplot2 package (Wickham, 2016) and Inkscape 1.0.1.

2.4 Experimental Design

We employed a between-subject experimental design, with environmental conditions (real and virtual) as the independent variables. A careful design process ensured that we could apply the experimental procedure in both environments with the smallest possible deviation. It was still necessary, however, to take into account the different hardware used and unpreventable effects under real conditions from the administrator or randomly intervening variables.

2.4.1 Experimental Procedure (Real Environment)

For the experiment in the RE, participants met the instructor in the university campus yard at the front of the building where the experiment was conducted. Participants signed an informed consent declaration and were given a small reward for participating (approx. 9\$). The experiment then commenced with participants obtaining the directions to Classroom A (and a paper with its number written on it). The participants were instructed to take the elevator to the fourth floor, but if they attempted to follow a different route, the instructor, walking behind at a distance, would stop them (Fig. 3). Participants walked the

route to Classroom A on their own accord without any intentional leading or prompting; this ensured the participants applied a wayfinding strategy and did not passively follow the administrator.

A second administrator, who was already waiting in Classroom A, calibrated the ET glasses with the participants when they arrived and read out the instructions for a distraction task (spotting ten differences in neutral photos which actually had only eight differences) and for evacuation (e.g., “do not touch the glasses, do not talk to the administrator”). The distraction task gave the second administrator enough time to prepare the action camera, which was used to record the participants from waist down as they attempted to evacuate from the building (participants were followed by the administrator). In the meantime, the first administrator locked the lattice doors which participants passed through on their way to Classroom A and put up a warning sign (“elevator cannot be used during evacuation”) on the elevator. The route, which the participants already knew, was thus blocked, and they were forced to find a new route themselves. After 90 seconds of the distraction task, the second administrator set an alarm off. It was loud enough to be heard on the entire fourth floor. If the participant did not react within ten seconds, the administrator would verbally announce that it was time to evacuate. After completing the evacuation phase, participants were informed of their success in completing the evacuation task.

2.4.2 Experimental Procedure (Virtual Environment)

For the experiment in the virtual environment, participants met the administrator in the laboratory (Virtual Geographic Environment laboratory - VGE Lab) where the virtual reality setup was located (Fig. 4). Participants signed an informed consent declaration and were given a small reward for participating (approx. 9\$). First, participants learned how to control the virtual environment in the Virtual Gallery training task. The experiment in the VE then commenced. At the start, the participants appeared at the front of the building and the administrator read out the directions to the classroom. If they attempted to use a different route, the administrator would tell them which direction they should go. Once the participants reached classroom A, the administrator calibrated the ET device (built into the headset). The administrator then revealed the distraction task (the same as in the RE) and then read out the instructions. Activating the fire alarm automatically barred certain doors and indicated a warning sign on the elevator. After finding their way out, participants were informed of their success in completing the evacuation task and were prompted to remove the VR headset.

2.5 Participants

The participants were recruited mainly from the population of students at Masaryk University or through social networks. Eighty-six participants were allocated into two groups: 39 for the real environment, 47 for the virtual environment. Some participants were excluded from the analysis (four from the RE and nine from the VE). The most common reason for exclusion was participants experiencing cybersickness in virtual reality. If a participant was susceptible to this phenomenon (sometimes even as they moved about in the practice virtual environment preceding the experimental virtual environment), the experiment was immediately stopped. Other participants were excluded because of eye tracking issues or bystander interference during the real scenario.

The experiment in the real environment involved 15 men and 20 women, with a mean age of 25.5 ($Mdn = 23.50$, $SD = 6.37$). The virtual environment involved 22 men and 16 women, with a mean age of 24.13 ($Mdn = 23.50$, $SD = 3.91$). Participants were only allowed to participate if they had no previous experience with the building and no serious vision or mobility impairment. Participants were informed about the research objectives and the evacuation simulation. They were also explicitly informed about their right to end their participation in the experiment at any time without losing the reward.

2.6 Measured Parameters

2.6.1 Locomotion

Participant movement in the virtual environment was captured automatically in virtual space coordinates at a rate of 120 times per second. In the real environment, we manually recorded participants from the waist down with a GoPro camera (e.g., Skorupka, 2009; Marín-Morales et al., 2019). Video records required manual post processing in ArcGIS Pro by entering estimates of the user's locations into a simplified 3D model of the building, at 15 FPS (i.e., at 500 ms intervals) with a corresponding timestamp assigned automatically.

The positional data of the virtual environment participants were first ex-post filtered as raw text for information relevant only to the study's aim (i.e., from the moment of the alarm to the eventual exit from the building). Each location was assigned to its respective floor according to its logged timestamp. The spatial granularity of the virtual environment data was also reduced to correspond more with data from the real environment.

This process ensured comparability between the datasets, but the different character of movement created an incomparability of net time and distance between the environments. In the real environment, walking speed was natural, continuous, and not limited whereas in the VE, participants moved at a constant speed (3 s/m). This constant speed was a consequence of the binary key control in the VE controller: a keyboard arrow is either on or off. We selected a constant speed of 3 m/s as a compromise between a natural walking speed of approximately 1.4 m/s (McNeill, 2002) and the much higher speeds of 4–10 m/s first-person gamers are generally accustomed to in most computer graphics experiences (Lee & Chang, 2017). This aided in balancing the experience for gamers and non-gamers alike.

From the processed datasets, we derived variables for further use in the comparative analyses. We extracted evidence of participants applying retracing strategies (binomial, indication of whether participants used the same exits to leave and enter the building or followed the evacuation signs instead), alarm reaction time (time required to take action and leave the room after the alarm sounded), decision time at doors (time required to address barred routes), and decision time at the crossing of evacuation routes (time required to choose the final section of the evacuation route). By localising clusters in point data, we identified the points where participants stopped during the evacuation, allowing us to evaluate the movement dynamics. Clusters were automatically identified by filtering: points were kept only from moments when an individual participant did not move away by more than X m for at least 2 seconds, where X corresponds to the average speed of all participants (i.e., 3 m/s set in the VE variant; calculated from the collected data, 1.5 m/s in the RE variant).

To obtain the participants' movement trajectories, we imported both processed datasets as points into ArcGIS Pro, where they were overlaid onto 2D schematic cross-sections of each floor. These points were then converted into trajectories by automatically linking subsequent points in time. An average trajectory was then calculated in GRASS GIS for each floor and condition. Several examples of a major deviation from other trajectories were not included, as they would significantly affect the resulting average evacuation trajectory. To assess the similarity or dissimilarity of the overall movement of the participants in both environments, we created concave hulls for each floor in QGIS from the trajectory lines and then overlaid and intersected them; major deviations were again excluded.

2.6.2 Eye Movement

The main aim with recording eye movements was to observe hit presence on key AOIs. If an AOI experienced a hit, it means that it was seen by the participant. This approach also removes the problem of not fully comparable eye-tracking data sets between the SMI mobile eye tracker and the eye tracker in the HMD. The fixation or intakes from these devices in the real environment were automatically calculated by the SMI BeGaze 3.7 program; we therefore cleared the data from the VE to correspond. All gazes on objects in the VE of less than 50 ms were deleted. The SMI manual states this value as the threshold value for intake recognition (SMI, 2017); this value is also widely accepted as the lowest for any possible cognitive recognition (Galley, Betz, & Biniossek, 2015). Eye behaviour was analysed in sequence from the beginning of the alarm to exiting the final door.

According to Sorrows and Hirtle (1999), orientation points can be either subjective or objective, and any object along a route can potentially serve as a person's orientation point. Many categories of landmarks have been established in the literature, and they partially overlap. Dichotomic categorization defines landmarks as objects or geometric landmarks (Gallistel, 1999). More categories based on visual accessibility are local and global landmarks. Global landmarks are visible from practically anywhere in the environment whereas local landmarks are unique to each part of the route (Hurlebaus, Basten, Mallot, & Wiener, 2008; Stankiewicz, B. J., & Kalia, A. A.). Ohm et al. (2014) used four categories (architecture, function, information, furniture); Viaene et al. (2016) defined their landmarks into 20 categories (fire, window, pictogram, written evacuation sign, area sign, evacuation, sign in a room, ornament, server, closet, radiator, poster, trash, light, fireplace, window, door route, door (other), staircase, various).

For the analysis in the current study, we kept most of the available objects. Our categorization was inspired by previous works, but we adjusted it to better fit our aims. These categories were defined to determine whether the evacuation orientation strategies differed in each environment. Navigation through a building can be practically accomplished in two ways: following signs or following a familiar route. In real emergencies, following a familiar route is the most common practice (Kinateder, Comunale & Warren, 2018; Sime, 1985). However, our participants were not sufficiently familiar with the building, and we did not expect the majority to retrace their paths during evacuation.

We also compared the extent of the participants' attention to (ir)relevant objects in both environments to discover whether the salience of these objects differed (Figs. 5 and 6). We categorised AOIs into three main categories:

1) Evacuation signage – these types of sign assist people in orienting themselves during an emergency. We further subdivided them into:

1a) *evacuation signs* – arrows, person on stairs signs, EXIT signs

1b) *evacuation plans* – for the current study, these were somewhat redundant since they were barely legible in the VE and improperly made in the RE and confused participants more than they assisted. It is important, however, to at least distinguish these two for the theory since a person's cognitive activity differs reading one or the other.

2) Common building elements – these objects are found in many buildings. We further subdivided them into:

2a) *furniture* (e.g., plants, tables, chairs, etc.)

2b) *information signs* to assist visitors in navigating through the building (e.g., class schedule, etc.).

3) Special functional elements – *windows*, for example, provide outdoor context and may assist in navigating.

2.6.3 Position - Gaze

Eye-tracking and movement data were possible to synchronise with the exact moment of the alarm sounding, thus allowing us to identify the position where a participant observed an object for the first time. These locations were processed in ArcGIS Pro as point data and overlaid onto 2D schematic cross-sections. Points for each AOI in both environments were grouped, and then concave hulls were created for each group. The concave hulls assigned to corresponding AOIs in both environments were then overlaid to determine the overlapping percentage. To compare the two environments, we focused primarily on evacuation signs, as those elements were positioned in exactly the same locations.

3. Results

3.1 Locomotion Analysis

For locomotion, we considered two sets of metrics. As the main global metrics, we observed evacuation time (starting from the participant's reaction to the fire alarm and exiting the classroom), the final distance walked, and the effect of the environment on the choice of escape door (retracing or following signs); as secondary metrics for analysis, we recorded the number and mean time of stops, reaction time to the alarm, decision time at barred doors, decision time choosing the escape door, and choice of the elevator (Tables 1 and 2).

Participants in the VE were quicker ($M = 86.47$ s, $SD = 15.42$) than participants in the RE ($M = 118.17$ s, $SD = 23.71$), indicating a statistically significant difference ($t = 6.825$, $df = 71$, $p < 0.001$, 95% CI [22.44, 40.96], $d = 1.43$). They also walked longer distances ($M = 183.91$, $SD = 19.76$) than the participants in the real environment ($M = 161.52$, $SD = 8.23$), also indicating a significant difference ($t = -6.22$, $df = 71$, $p < 0.001$, 95% CI [-29.56, 15.21], $d = 1.46$). The choice of final escape route indicated no significant statistical difference between the participants ($p = 0.226$, 95% CI [0.62, 5.85], $OR = 1.87$, Fisher's exact test); the majority of participants followed the evacuation signs: in the RE, 26 participants (= 74%, 9 participants retracing) followed the signs, and in the VR, 23 (= 60.5%, 15 participants retracing).

Table 1
Analysis of secondary numeric locomotion data

Comparison	Condition	Mean	SD	t	df	p	95% CI	d
Number of stops	VE	8.12	2.83	-10.136	71	<0.001*	2.714, 8.132	2.375
	RE	2.71	1.47					
Average duration of stops	VE	4.56	1.38	0.542	71	0.5898	-1.199, 2.094	0.127
	RE	5.01	4.89					
Reaction time on alarm	VE	6.91	3.18	-1.5658	71	0.1218	-0.609, 0.314	0.366
	RE	5.76	2.19					
Decision time at barred doors	VE	8.62	7.86	1.1468	71	0.2553	-1.959, 7.266	0.269
	RE	11.27	11.68					
Decision time before choosing escape door	VE	1.79	1.09	2.81	71	0.0064*	0.361, 2.123	0.658
	RE	3.03	2.47					
* Statistically significant results								

Table 2
Analysis of secondary categorical locomotion data

Comparison	Condition	Count YES (%)	Count NO (%)	p	95% CI	OR
Using the elevator	VE	4 (10.5)	34 (89.5)	0.748	0.256, 22.623	1.92
	RE	2 (5.7)	33 (94.3)			

In addition to the statistical analysis, we performed an exploratory data analysis using GIS software. The average trajectories on the fourth floor (Fig. 7) were, for the most part, similar. Discrepancies were evident in the staircase areas of the fourth, third and second floors (Figs. 7–9.), where participants in the RE tended to follow the inner railings of the staircase in contrast to the participants in the VE, who walked more along the centre-line of the staircase. By contrast, in some of the areas between individual staircases, participants in the VE stayed closer to the staircase's columns. This trend of staying closer to corners/centre was also somewhat evident on the first floor (Fig. 10). In choosing the main exit as an evacuation route, some participants in both environments first hesitated and opted to retrace to the side exit, but then decided on the main exit. In this case, the average trajectories did not indicate any major differences. The opposite was evident from the average trajectories of participants who chose the side exit as an evacuation route, where we can see considerable differences. These, however, may have been partially caused by the door opening mechanics in the VE. It is also interesting that VE participants, who tended to become lost more often, demonstrated a greater number of deviations from the main body of trajectories, whether as a result of disorientation or simply exploring the environment.

Examining the similarity of the concave hulls for all trajectories in both environments, we calculated the overall overlap for the entire evacuation as approximately 70% (69.9% for the side exit route and 68.8% for the main exit route); the overlap values for each floor are depicted in the floor plans in Figs. 5–8.

The estimated mean sinuosity for the real environment and the virtual environment were 4.76 and 5.04, respectively. These values suggest that movement through the VE was less linear, perhaps as a consequence of the control interface design or computer peripherals setup and the inexperience of participants with virtual reality. Ultimately, this is evident from the floor plans, where the majority of non-overlapping areas consisted of trajectories from the VE.

3.2 Analysis of Eye Movements

All primary selected RE participants were included in the analysis since the cumulative data loss from all measurements did not exceed 8%. We also visually checked the accuracy of the eye-tracking data and detected no problems. One VE participant was excluded from the analysis since his data loss exceeded 8%.

The main result from this analysis is whether an object in the environment (AOI) was seen. We also modelled the distance from the AOI for the moment when participants saw the object for the first time. To model the binary dependent variable we used a generalised linear mixed model with a logit link; to model the distance, we used a linear mixed model. We applied the lme4 package (Bates et al., 2015) in R to fit both types of model. The level-1 unit of analysis was a recorded encounter of a participant with an AOI, with random effects for participants and AOIs accounting for the unmodeled differences between participants and the AOIs.

The predictor of primary interest is whether the evacuation took place in the RE condition (coded 0) or in the VE (coded 1). The individual tendency to notice specific AOIs is a factor we want to control for; individuals are therefore specified as a random factor in our models. However, it should be noted this factor is partially confounded by the VE/RE condition as due to the between-subjects design.

Each AOI category contained multiple individual elements, i.e., evacuation signs (30), evacuation plans (4), information signs (7), windows (23) and furniture (24), and we modelled each category separately. We did not consider four evacuation plans a sufficient sample; additionally, they were difficult to read, and we therefore chose not to run an analysis for this category. For the purposes of the analysis, we considered individual AOIs which fell into the above categories as random samples of that particular category. This assumption was somewhat unrealistic since at least some of the individual AOIs may have had specific potential to assist with evacuation. Our sample, however, was too low to treat them as fixed, and consequently the focus of the analysis was more general. Again, our intention was to statistically measure the differences in conspicuousness of individual AOIs and their potential in better estimating the effects of the VE on their use.

In separate models, we statistically controlled for several other factors and covariates which potentially conflated the effects of the VE. The predictors included corrected runtime, gender, whether the participant attempted to use the elevator, and the floor where the AOI was located. In respect of the sample size, the observed effects were unstable, small and mostly non-significant. The effect of the VE was smaller after including these predictors. Extended models are reported in the online supplement.

3.2.1 Visibility Models for Individual AOI Categories

Using generalised linear mixed models with logit links estimated by glmer, we modelled the dichotomous visibility (Bates et al., 2015). For each AOI category, we report two models: (1) a baseline, which includes an intercept and two random effects (AOIs and participant IDs) and (2) a model which includes the environment condition as both a fixed factor and a random factor. The covariance between the random intercepts and random effects of the VE on AOIs was also modelled. Table 3 reports the parameters, fit and predicted success rates of all eight models.

The raw probability of seeing an evacuation sign was 22.4% (490 of the 2190 opportunities to visually rest on an AOI were successful). A small raw difference in the probability of seeing a sign is apparent (24.7% in the VE and 19.8% in the RE). The baseline model has an overall 83% success rate (95% true positives, 43% true negatives). Both random effects are substantial and statistically significant, with large differences between the participants seeing the AOIs and the AOIs being seen. The intercept, which represents the predicted probability of an average participant seeing an average evacuation sign, is estimated at -1.95, which is equivalent to a probability of 0.12.

When added to the model, the fixed effect of the VE is very small ($b = .23$, $p = .632$). The odds of an average person seeing an average evacuation sign is 1.25 times greater in the VE (95% CI for OR = (.50, 3.15)) which, translated into probabilities, is 9% in the RE versus 11% in the VE. It must be noted that the confidence interval is very wide, implying that the fixed effect is non-significant; we cannot be certain that it is low, however. Interestingly, the random effect of the VE for evacuation signs is fairly large (SD of the effect of VE is 1.69 (95% CI (1.1, 2.2))), indicating that some signs were more visible in the VE than others. The model with the VE shows a better fit to the data in terms of AIC and BIC.

The raw probability of seeing an information sign is 44.4%. The difference between the RE and VE is also small, at 42% and 47%, respectively. The baseline model has an overall 81% success rate. Both random effects are substantial and statistically significant, with large differences between the participants seeing the AOIs and the AOIs being seen.

When added to the model, the fixed effect of the VE is very small ($b = .22$, $p = .557$). The odds of an average person seeing an average information sign is 1.25 times greater in the VE (95% CI for OR = (.60, 2.61)), which, translated into probabilities, is 37% in the RE versus 42% in the VE. In this case, the confidence interval is also wide, implying that the fixed effect is non-significant, but we cannot be certain it is low. The random effect of the VE for information signs is low although still statistically significant (SD of the effect of VE is .59 (95% CI (.04, 1.2))). Overall, adding the VE to the model did not improve its fit, where both AIC and BIC increased.

The raw probability of seeing (through) a window is 20.0%. The difference between the RE and VE is also small, at 17% and 23% respectively. The baseline model has an overall 82% success rate. Both random effects are substantial and statistically significant, but the differences between the participants seeing the AOIs and the AOIs being seen are smaller than those in both types of sign.

When added to the model, the fixed effect of the VE is small but statistically significant ($b = .44$, $p = .023$). The odds of an average person seeing an average window is 1.56 times greater in the VE (95% CI for OR = (1.06, 2.28)), which, translated into probabilities, is 13% in the RE versus 19% in the VE. The random effect of the VE for windows could not be estimated because of estimation problems. Overall, adding VR to the model improved its fit in terms of both AIC and BIC, but only marginally, which we can infer from the unchanged prediction success rate.

The raw probability of seeing an item of furniture is 39.3%. The difference between the RE and VE is also small, at 42% and 37%, respectively. The baseline model has an overall 71% success rate. Both random effects are substantial and statistically significant, but the differences between the participants seeing the AOIs and the AOIs being seen are smaller than those in both types of sign.

When added to the model, the fixed effect of the VE is very small ($b = -.36$, $p = .299$). The odds of an average person seeing an average item of furniture is 1.42 times lower in the VE (95% CI for OR = (.36, 1.37)), which, translated into probabilities, is 41% in the RE versus 32% in the VE. The random effect of the VE for items of furniture is fairly large (SD of the effect of VR is 1.44 (95% CI (0.89, 1.90))), indicating that some pieces were more visible in the VE than others. Overall, adding VR to the model improved its fit in terms of both AIC and BIC and a slightly increased overall prediction success rate.

Table 3

Parameters of the logistic mixed models of visibility of the four object categories in the evacuation environment

	<i>Evacuation signs</i>					<i>Information signs</i>					<i>Windows</i>					<i>Furniture</i>	
	Baseline		Effect of VR			Baseline		Effect of VR			Baseline		Effect of VR			Baseline	
<i>Fixed effects</i>	b	SE(b)	b	SE(b)	OR (95% CI)	b	SE(b)	b	SE(b)	OR (95% CI)	b	SE(b)	b	SE(b)	OR (95% CI)	b	SE(b)
<i>Intercept</i>	-1.95	0.31	-2.29	0.40	0.10 (0.05; 0.22)	-0.43	0.62	-0.55	0.61	0.58 (0.17; 1.93)	-1.68	0.21	-1.91	0.23	0.15 (0.09; 0.23)	-0.51	0.2
VR			0.23	0.47	1.25 (0.50; 3.15)			0.22	0.38	1.25 (0.60; 2.61)			0.44	0.19	1.56 (1.06; 2.28)		
<i>Random effects</i>																	
<i>Intercept (ID) - SD</i>	1.17		1.27			0.81		0.82			0.64		0.60			0.53	
<i>Intercept (AOI) - SD</i>	1.45		1.70			1.59		1.52			0.85		0.85			0.87	
VR (AOI) - SD			1.69					0.59					–				
r (Intercept. VR)			-0.35					0.23					–				
<i>Fit</i>																	
-2LL	1848		1777			545		542			1532		1527			2132	
AIC	1854		1789			551		554			1538		1535			2138	
BIC	1871		1824			564		579			1554		1557			2155	
<i>Success rate</i>																	
overall	83%		85%			81%		82%			82%		82%			71%	
positive	43%		49%			84%		84%			18%		18%			45%	
negative	95%		95%			79%		81%			98%		98%			87%	

3.2.2 Gaze - Position Analysis

For objects which guaranteed a precisely equivalent position in the real and virtual environments, we were able to analyse the participant's position and distance from the object he or she observed for the first time.

Using linear mixed models estimated by lmer, we were able to model the distance (in metres) of the participant from an object in the evacuation environment (AOIs) at the time of the participant's first fixation on the object (Bates et al., 2015). For each AOI category, we report two models: (1) a baseline, which includes an intercept and two random individual AOI effects only and the participant's ID; (2) a model which includes the evacuation environment (RE = 0, VE = 1) as both a fixed factor and a random factor affecting the AOIs. The covariance between the random intercepts and random effects of the VE on AOIs was also modelled. Table 4 presents descriptive statistics for the distances. Table 5 lists the parameters, explained variances and intra-class correlations of all eight models.

Table 4
Descriptive statistics for the distances at which the AOsI were first seen

	condition	n	M	SD	min	max	skew	kurtosis
Evacuation signs	VE	282	5.94	2.98	0.52	19.31	0.53	0.74
	RE	208	5.72	3.93	0.87	25.80	2.10	6.58
Information signs	VE	124	6.93	2.08	2.15	15.29	0.03	0.92
	RE	103	6.30	2.34	1.60	10.11	-0.08	-1.38
Windows	VE	200	6.42	3.42	1.28	17.47	0.67	0.01
	RE	136	5.07	3.21	0.92	18.38	1.18	1.50
Furniture	VE	333	10.43	6.87	0.57	43.38	2.35	6.66
	RE	356	9.89	6.38	1.24	45.65	2.54	9.25
Note: The values of n are smaller in this table; only objects which were seen have been included in the analysis.								

According to the model, an average evacuation sign was first noticed by an average participant from a distance of 5.2 metres (bootstrap 95% CI (4.3,6.0)). A fairly small (only 3% of total variance) and non-significant variability between individual participants is evident (SD = 0.61 m). The variability between signs, however, is substantial (SD = 2.1 m) and accounts for 35% of the variance in distances. Adding the fixed and random effects of the VE increased the total explained variance from 38–45%, but it is mainly due to the large random effect of the VE for signs (SD = 1.7 m). The mean effect of the VE is miniscule ($b = -0.2$ m, $p = .747$).

The parameters of the model for information signs are very similar to the model for evacuation signs, the only difference being a higher variability between participants (SD = 0.96 m, 95% CI (0.60, 1.29)). Even though the estimated difference in distance is higher at 0.88 m, the effect is non-significant ($p = .192$).

For windows, the effect of the VE is larger; the average window is first seen by an average person from a distance of 1.6 m ($p < .001$), which is greater than in the RE.

The model for furniture is slightly different from the previous models. The distances were highly positively skewed and led us to modelling the natural log-transformed distance, consequently increasing the difficulty of interpreting the parameters. The pattern is nevertheless analogous to the models of both categories of signs. The fixed effect of the VE is nearly 0, with all variance due to the random effects accounted for by the model.

In summary, the distance at which the objects in the environment were first seen (if seen at all) was roughly the same in the VE as the RE, except for windows, which were seen sooner at a distance of approximately 1.6 m.

Table 5
Parameters of the linear mixed models of the distances of seeing the four object categories in the evacuation environment

	<i>Evacuation signs</i>				<i>Information signs</i>				<i>Windows</i>				<i>Furniture*</i>			
	Baseline		Effect of VR		Baseline		Effect of VR		Baseline		Effect of VR		Baseline		Effect of VR	
<i>Fixed effects</i>	b	SE(b)	b	SE(b)	b	SE(b)	b	SE(b)	b	SE(b)	B	SE(b)	b	SE(b)	b	SE(b)
<i>Intercept</i>	5.18	0.43	5.20	0.50	6.24	0.47	5.77	0.58	5.15	0.52	4.28	0.49	2.11	0.07	2.08	0.07
VR			-0.16	0.50			0.88	0.53			1.56	0.31			0.08	0.08
<i>Random effects</i>																
<i>Int</i> (ID) - SD	0.61		0.73		0.98		0.96		0.98		0.46		0.16		0.15	
<i>Int</i> (AOI) - SD	2.05		2.14		1.13		1.35		2.29		2.07		0.33		0.29	
VR (AOI) - SD			1.66				1.01				0.59				0.28	
<i>r</i> (<i>Int</i> . VR)			-0.34				-0.51				0.67				0.03	
Residual VAR	2.74		2.56		1.68		1.64		2.20		2.17		0.46		0.45	
<i>Fit</i>																
Conditional R2	.380		.447		.441		.505		.562		.569		.379		.424	
ICC(ID)	.031				.190				.087				.069			
ICC(AOI)	.349				.251				.476				.310			
<i>Notes.</i> *For furniture, the natural logarithm of the distance was modelled. Parameters in bold for the effect of VR are $p < .05$. VR - virtual reality condition coded 1; real condition coded 0.																

In the exploratory analysis, we selected the overlap of the concave hulls on the positions of first visits with respect to evacuation signage in the RE and VE to investigate the spatial distribution of first gaze visits (Figs. 11–13). The concave hull overlap of all objects varied considerably, ranging from 9.77–52.97%. In corridors, the participants in the VE demonstrably observed the evacuation signage from a greater distance than the participants in the RE. On staircases, however, the distances were more disbalanced. In most cases, the distance between the farthest visit and the nearest first visit along the evacuation route for the RE participants was greater. The data for the third floor could not be visualised since the first observation points for any of the evacuation signs from which an overlap could have been plotted were insufficient (the bare minimum was 3).

4. Discussion And Conclusion

We compared human behaviour during a simulated evacuation in a real environment and a virtual environment. Because these types of studies (where the virtual environment is directly comparable to the real environment) are currently few, we applied many different approaches to compare the conditions and found that both environments are comparable in all important aspects.

Regarding locomotion, we found an approximately 70% trajectory overlap, with differences mainly on the stairs, where participants in the real building tended to stay closer to the railings and walked around corners at a greater distance. These differences can probably be explained by safety measures which apply in the real world. The most significant differences in locomotion were found in movement fluidity. In the VE, people moved more quickly, stopped more often, and walked longer distances than those in the RE. They also spent less time deciding which exit to use, but the critical factor, namely the final choice, was comparable in both environments. The differences we found in the locomotion analysis might be explained by the character of the VE itself. Walking speed is dependent on the virtual environment setting. The quicker than usual speed with no tactile feedback from the participant's legs and the greater fluidity in locomotion may have propagated the longer distances walked in the VE since a virtual body is easier and quicker to set in motion. In the visual orientation analysis, we found that both groups were equally sign-dependent. Other AOI categories were also equal, with the exception of windows, which were fixated on more in the VE. Windows in the VE were also first seen from a greater distance, this being the only difference regarding distance and noticing the AOIs. We also found intergroup differences, mainly for signs and furniture, suggesting that not all objects along the route were equally important to the participants. More specifically, statistical analysis revealed that the differences caused by the individual differences and visual-spatial qualities of the objects (AOIs) were distinctly more significant than any effect from the environment (RE vs. VE).

The unique nature of our experiment means it is challenging to compare the results with other studies. Previous research has shown no difference in the ability to generally navigate and comprehend space in a virtual or real environment (Pastel et al., 2020; Pastel et al. 2021; Dong et al., 2021). This could be partially compared to exit route choice in our research since it depends on the realisation of self within the virtual building. Drewes, Feder and Einhäuser (2021) also used a replica of an existing building and found no differences in visual orientation, even though they used a less detailed model (e.g., no light rendering). Research by Dong et al. (2021) produced very similar results in an experiment where the participants searched for various rooms in a building. Both of the aforementioned studies also reported differences between sections of the route, more so than differences between the environments. As yet, these results are consistent with our findings.

We confirmed that virtual reality is a suitable tool for simulating evacuation and wayfinding scenarios. Behaviour exhibited by people was comparable in every significant way, with small differences only in locomotion, probably due to the artificial movement method in the VE. The experiment emphasised a high level of realism, which is an important factor in comparing environments (Lokka et al., 2018). The research design could benefit by, considering its aims, focusing less on the graphic aspects and using more advanced movement controllers as another functional approach (Drewes, Feder & Einhäuser, 2021). The artificiality of walking in a virtual environment could be lessened with the use of a treadmill or similar device, but the question of the best movement control remains open. Many options can be applied, for example oscillating the controllers to affect the speed of walk or allowing the participants to genuinely walk (treadmill or NaviBoard solution) (Hashemian et al., 2021). Because our aim was to determine whether it is possible to compare behaviour without any supplemental devices, walking speed could be reduced in a similar research design to be more human-like, or key-dependent speed change (as seen in video games) could be implemented.

Declarations

Compliance with Ethical Standards

All participants signed an informed consent. They were also informed verbally about the procedure and about the possibility to terminate the experiment at any time without loss of the right to remuneration.

Competing Interests

The authors have no competing interests to declare that are relevant to the content of this article

Data availability

All data generated or analysed during this study are included in this published article [and its supplementary information files].

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Figures

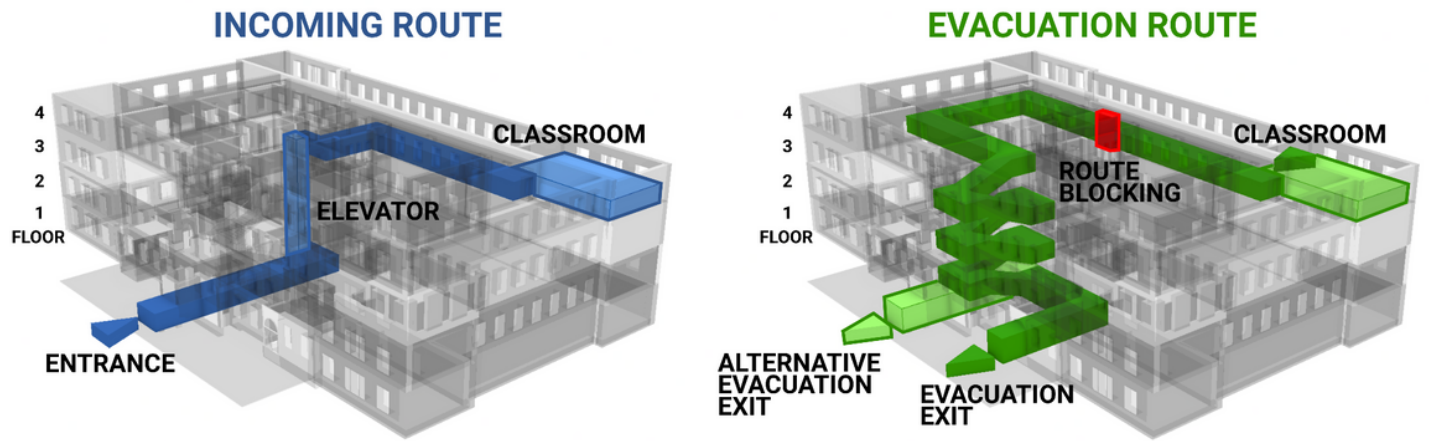


Figure 1

Overview of the experimental environment with marked routes



Figure 2

Comparison of real (left) and virtual (right) environments

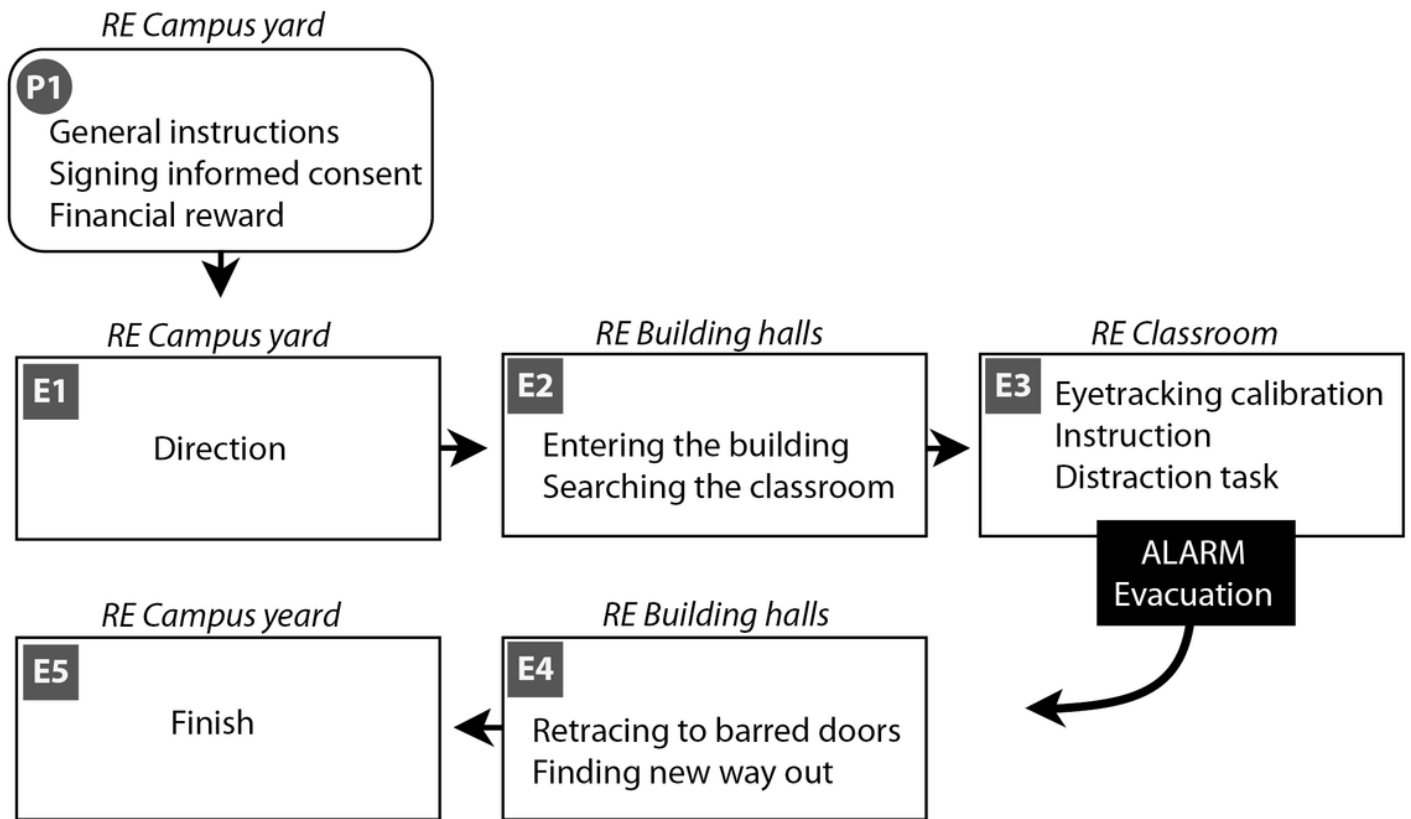


Figure 3

Experimental procedure for the real environment

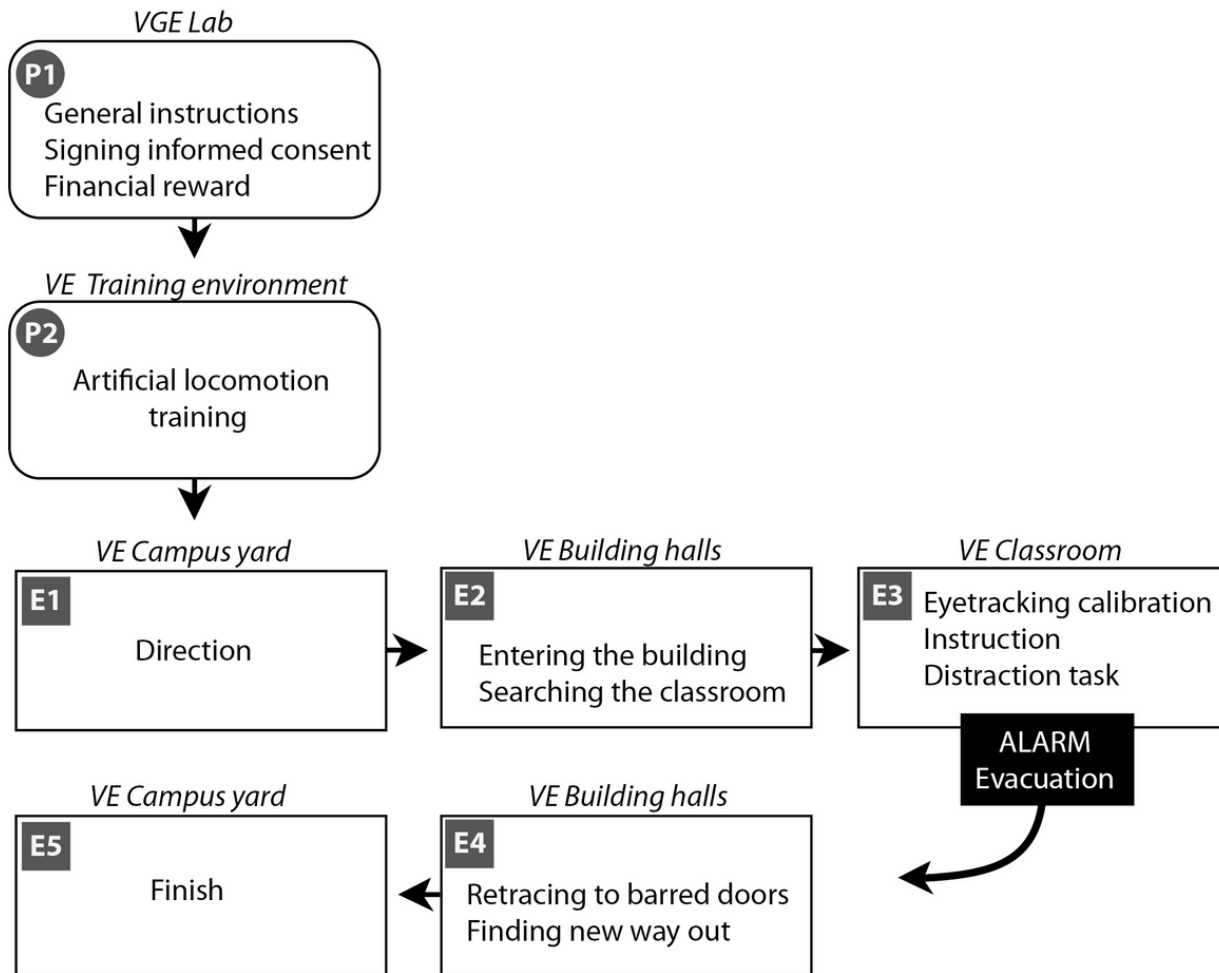


Figure 4

Experimental procedure for the virtual environment

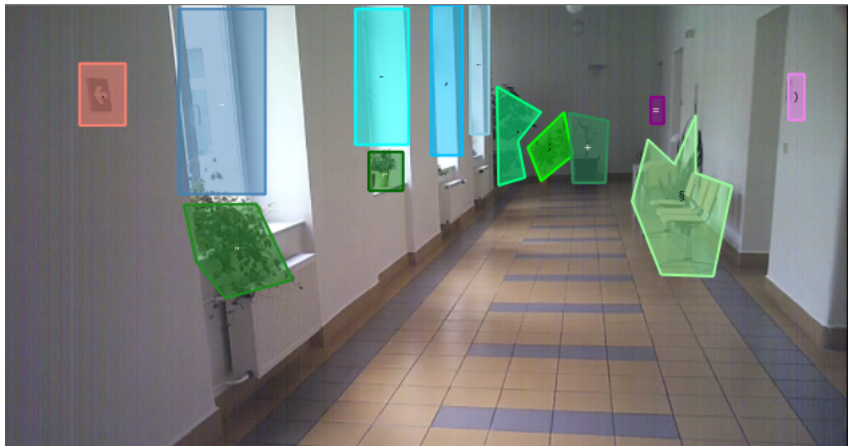


Figure 5

Examples of AOIs and corresponding categories in the RE (green = furniture (2a), blue = windows (3), orange = evacuation sign (1a), pink = information signs (2b))



Figure 6

Examples of AOIs in the VR (left = evacuation sign (1a), right = furniture (2a))

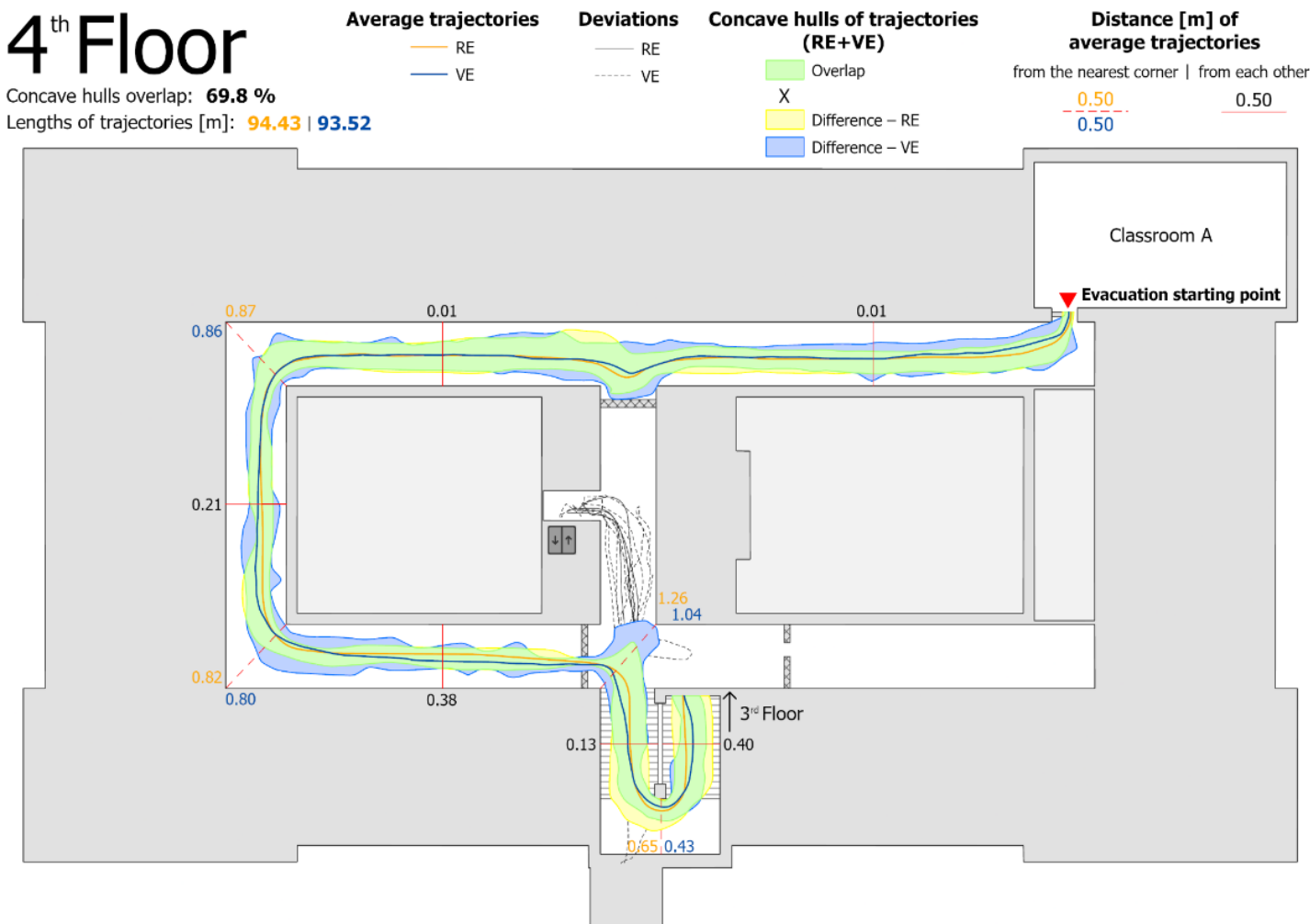


Figure 7

Visualisation of average evacuation trajectories, concave hulls of trajectories and selected metrics for both environments (fourth floor)

3rd Floor

Concave hulls overlap: **69.3 %**
 Lengths of trajectories [m]: **17.87** | **18.61**

Average trajectories

RE

VE

Deviations

VE

Concave hulls of trajectories (RE+VE)

Overlap

X

Difference - RE

Difference - VE

Distance [m] of average trajectories

from the nearest corner

from each other

0.50

0.50

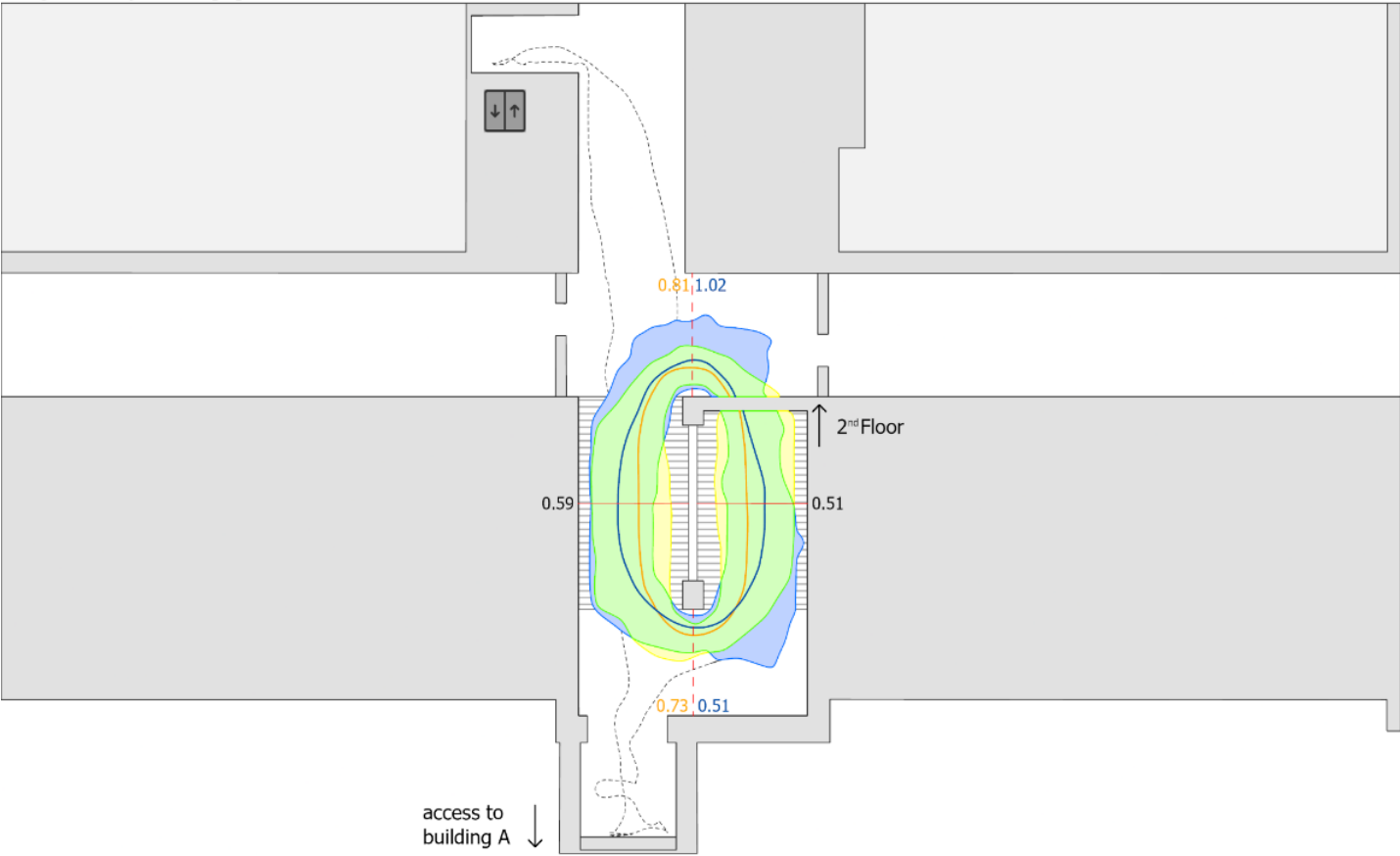


Figure 8
 Visualisation of average evacuation trajectories, concave hulls of trajectories and selected metrics for both environments (third floor)

2nd Floor

Concave hulls overlap: **72.7 %**

Lengths of trajectories [m]: **17.97** | **18.20**

Average trajectories

RE

VE

Deviations

VE

Concave hulls of trajectories (RE+VE)

Overlap

X

Difference – RE

Difference – VE

Distance [m] of
average trajectories

from the nearest corner | from each other

0.50

0.50

0.50

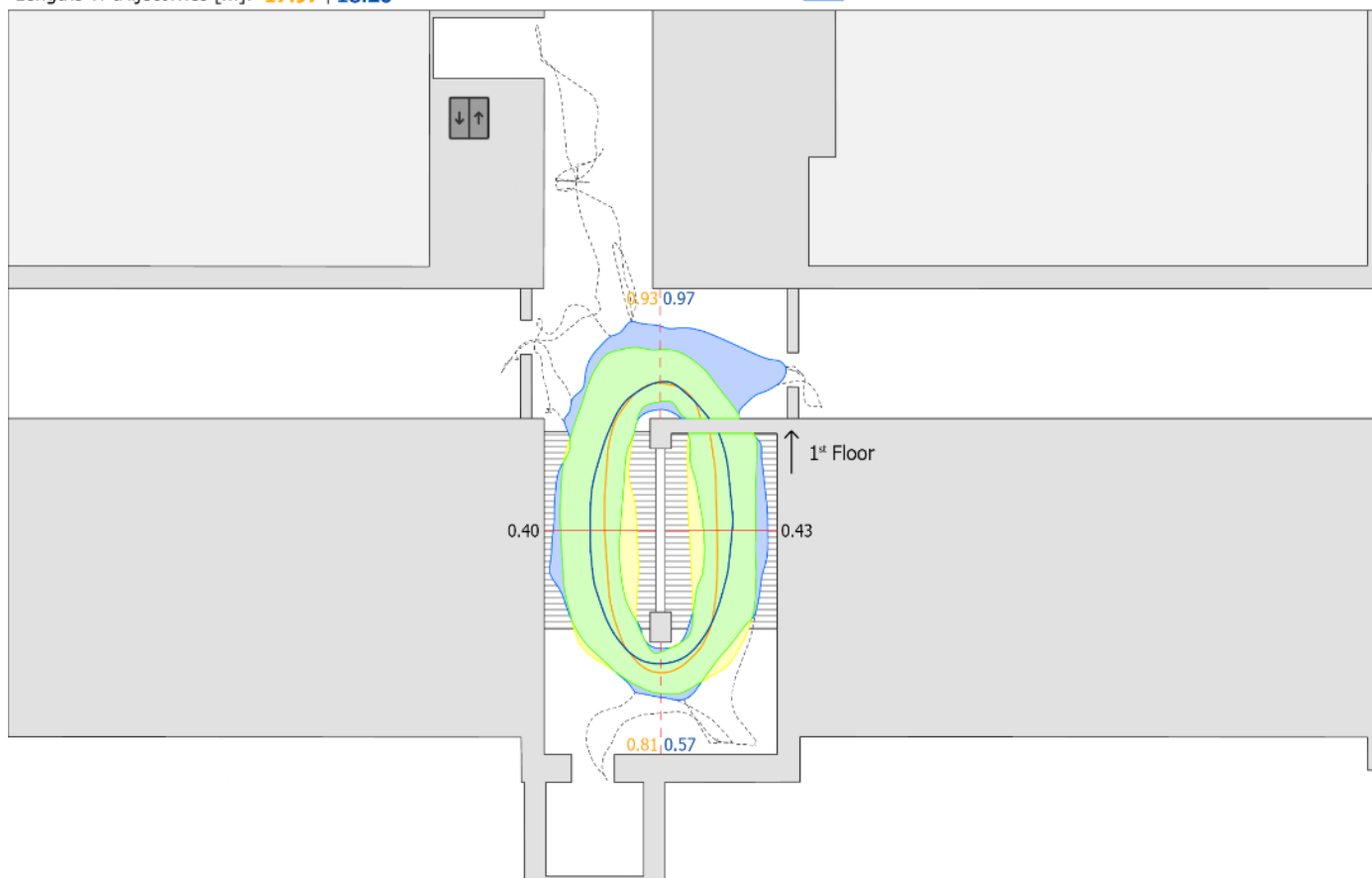


Figure 9

Visualisation of average evacuation trajectories, concave hulls of trajectories and selected metrics for both environments (second floor)

1st Floor

Concave hulls overlap:

56.3 % (side exit)

68.1 % (main exit)

Lengths of trajectories [m]:

side exit

16.39 | 16.75

main exit

22.84 | 22.97

Average trajectories

— RE

— VE

Deviations

— RE

--- VE

Concave hulls of trajectories

(RE+VE)

Overlap

X

Difference – RE

Difference – VE

Distance [m] of
average trajectories

from the nearest corner | from each other

0.50

0.50

0.50

0.50

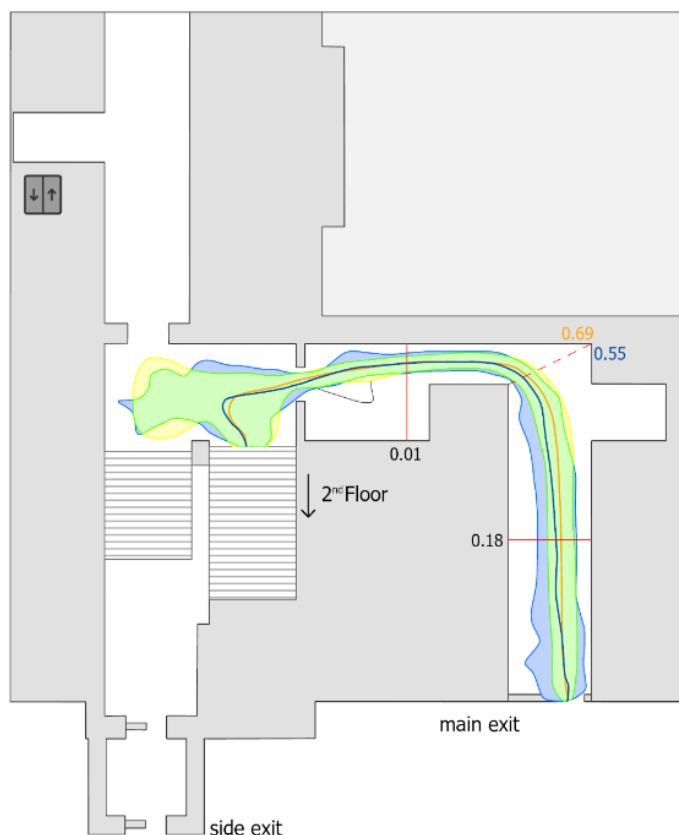
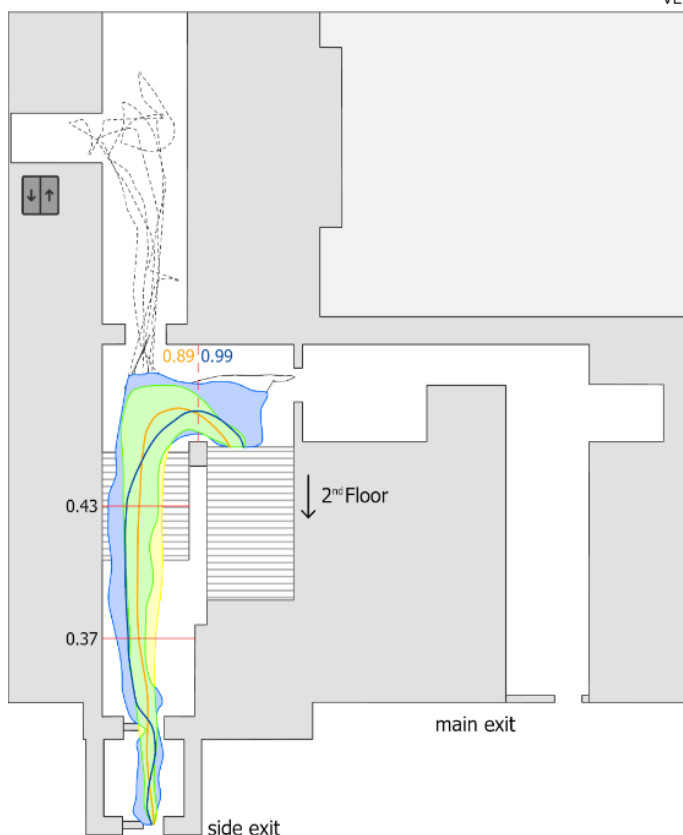
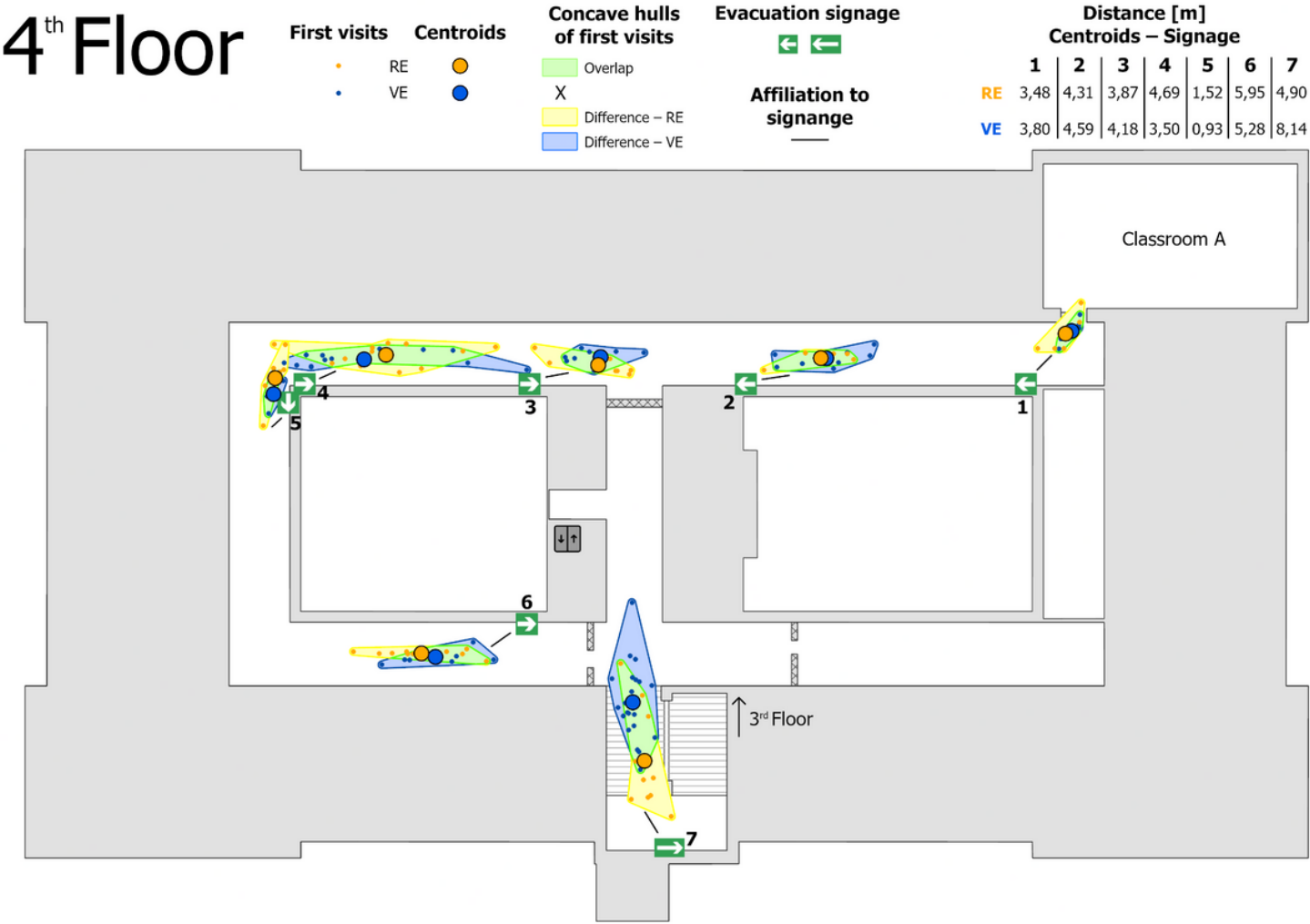


Figure 10

Visualisation of average evacuation trajectories, concave hulls of trajectories and selected metrics for both environments (first floor)



2nd Floor

First visits Centroids

• RE •
• VE •

Concave hulls
of first visits

Overlap
X
Difference – RE
Difference – VE

Evacuation signage



Affiliation to
signage

Distance [m]
Centroids – Signage

	8	9	10
RE	5,89	4,62	3,70
VE	6,28	4,59	3,10

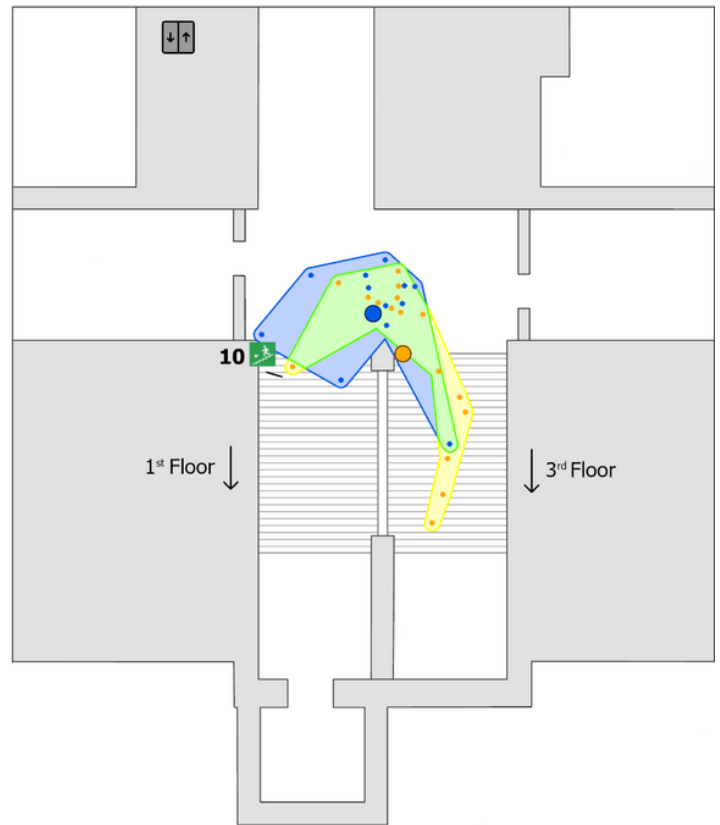
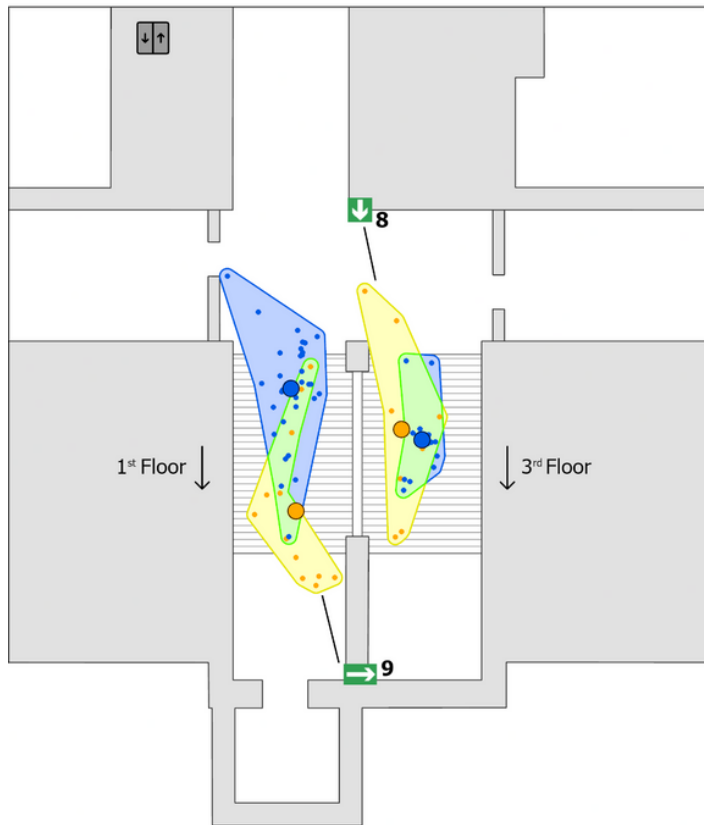


Figure 12

Visualisation of the spatial distribution of first visits, concave hulls of first visits, and percentage overlaps of these concave hulls (second floor)

1st Floor

First visits Centroids

• RE •
• VE •

Concave hulls
of first visits

Overlap
X
Difference – RE
Difference – VE

Evacuation signage



Affiliation to
signage

Distance [m]
Centroids – Signage

	15	16	17	18
RE	3,93	6,20	8,50	3,26
VE	3,51	8,66	8,18	3,82

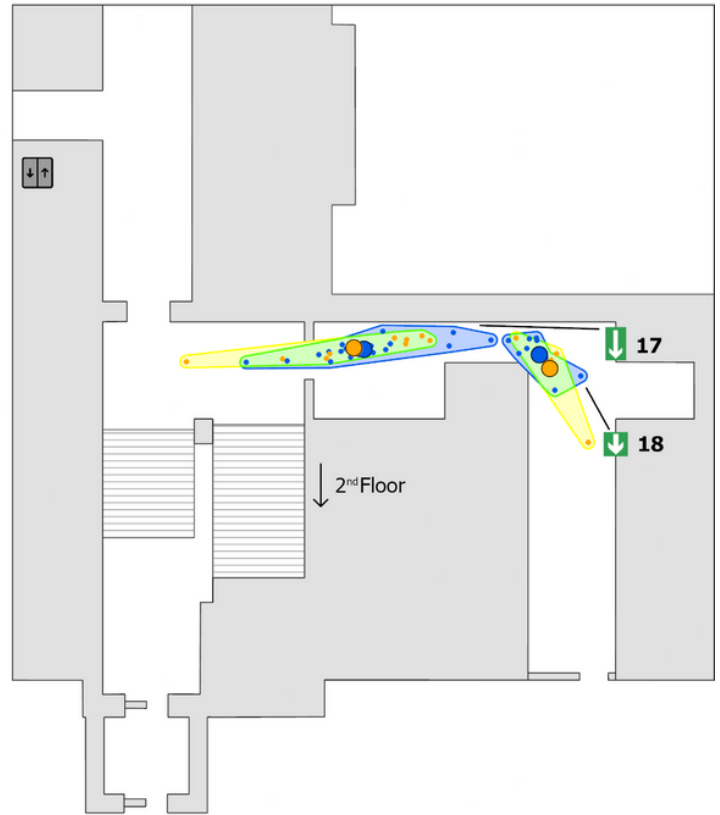
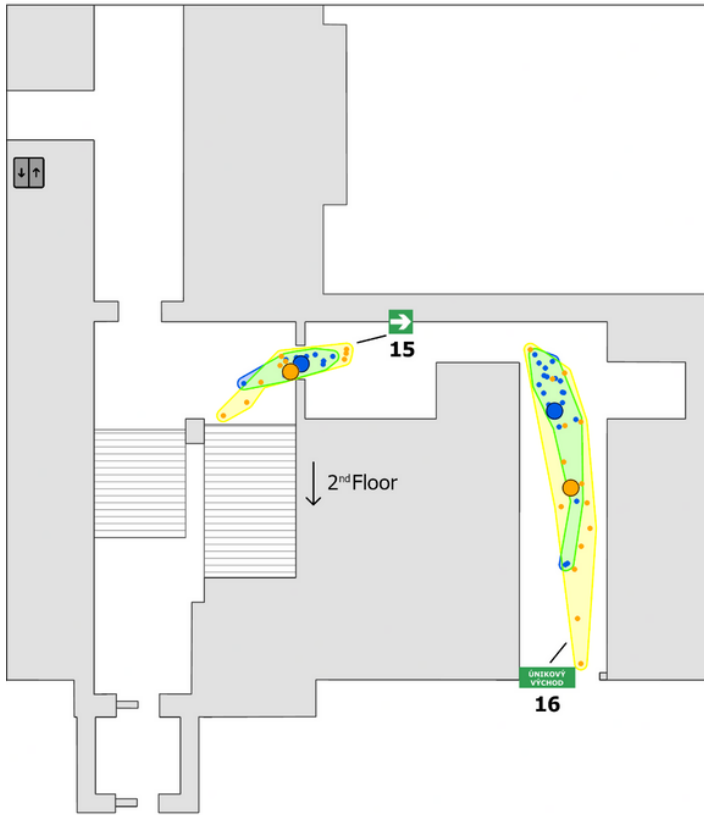


Figure 13

Visualisation of the spatial distribution of first visits, concave hulls of first visits, and percentage overlaps of these concave hulls (first floor)